

HygieneBench: A Reproducible Synthetic Benchmark for Cyber-Hygiene Anomaly Detection Across Identity, Endpoint, and Patch Telemetry

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Abstract—Cyber-hygiene anomaly detection, identifying stale privileged accounts, dormant-account reactivations, endpoint-agent coverage gaps, patch noncompliance clusters, and telemetry missingness, is operationally important but poorly served by existing public benchmarks, which focus predominantly on network intrusion or attack-emulation telemetry. We introduce HYGIENEBENCH, an open, reproducible benchmark that jointly covers Active Directory identity state, endpoint patch posture, vulnerability exposure, and telemetry freshness across seven evaluation tasks (T1-T7) and twelve anomaly classes. All telemetry is generated from a seeded synthetic pipeline grounded in public structural priors (NIST NVD severity distributions, Verizon DBIR 2026 patch-lag aggregates, CISA BOD 23-01 cadence requirements). We evaluate eight detection methods under five telemetry conditions (baseline, fresh, stale, missing, unsupervised) across 810 runs using a pre-registered failure protocol. Our central finding is that 83.8% of (condition, task, method) configurations fail to consistently outperform a structured rule baseline by the pre-registered $\Delta \geq 0.05$ AP threshold. ML adds meaningful signal on group-membership-drift detection (T2, best ML: temporal z-score, +0.185 AP) and patch-vulnerability hygiene (T5, best ML: OCSVM, +0.210 AP). Generator, datasets, and evaluation harness are openly released.

Index Terms—cyber hygiene, anomaly detection, benchmark, synthetic data, Active Directory, endpoint security, negative-result reporting

I. INTRODUCTION

Security operations centers (SOCs) in public-sector environments face persistent cyber-hygiene failures: stale privileged accounts invisible to behavioral analytics, endpoints lacking agent coverage, unpatched vulnerabilities aged well beyond CISA Binding Operational Directive 23-01 deadlines [1], and telemetry gaps that silently degrade detection confidence. NIST SP 800-40 Rev. 4 [2] frames enterprise patch management as a risk-based discipline requiring current, accurate asset and patch telemetry. Hygiene-state monitoring is a prerequisite for downstream identity and endpoint detection, without knowing which accounts are dormant, which agents are missing, and which patches are overdue, higher-order anomalies cannot be reliably attributed.

Yet no public benchmark exists for this problem class. Existing collections, the LANL Unified Host and Network Dataset [3], the CERT Insider Threat Dataset [4], CICIDS [5], and attack-emulation repositories [6], provide rich telemetry for

network intrusion or insider-threat detection but do not model identity hygiene state, patch posture, vulnerability exposure, or telemetry freshness as first-class evaluation axes.

This paper makes the following contributions:

- C1 A fully open, seeded synthetic telemetry generator covering eleven entity/event tables with documented structural priors.
- C2 A twelve-class cyber-hygiene anomaly taxonomy with ATT&CK enabling-condition mappings.
- C3 Seven benchmark evaluation tasks under five telemetry conditions with pre-registered evaluation budgets.
- C4 An 810-run comparative evaluation of eight detection methods with a pre-registered failure protocol.
- C5 Explicit negative-result reporting for 83.8% of configurations, with analysis of when and why ML fails to add value.

II. RELATED WORK

Intrusion and attack-emulation benchmarks. LANL [3], CICIDS [5], and DARPA TC [7] target network-level events. Mordor/Security Datasets [6] provides attack-emulation telemetry from red-team exercises. None models hygiene state or telemetry freshness as controlled evaluation variables.

Insider threat datasets. The CERT dataset [4] covers user behavior for insider-threat detection but does not include patch posture, vulnerability records, or the multi-table structure characteristic of enterprise AD environments.

Graph anomaly detection. Attributed-graph anomaly detection has produced several methods and datasets, including DOMINANT [8] and PyGOD [9], which operate on single-hop graph structure and do not model the temporal, multi-table telemetry of enterprise security operations.

Synthetic generation. Synthetic dataset generation has precedent in network simulation [10]. HYGIENEBENCH extends this tradition to the hygiene-state domain with eleven entity/event tables and publicly citable structural priors (NIST SP 800-40 [2], CISA BOD 23-01 [1]).

Negative-result reporting. Explicitly reporting when ML fails to outperform simpler baselines is established in machine learning [11] but underrepresented in security benchmarks. HYGIENEBENCH operationalizes this with a pre-registered Δ -threshold protocol.

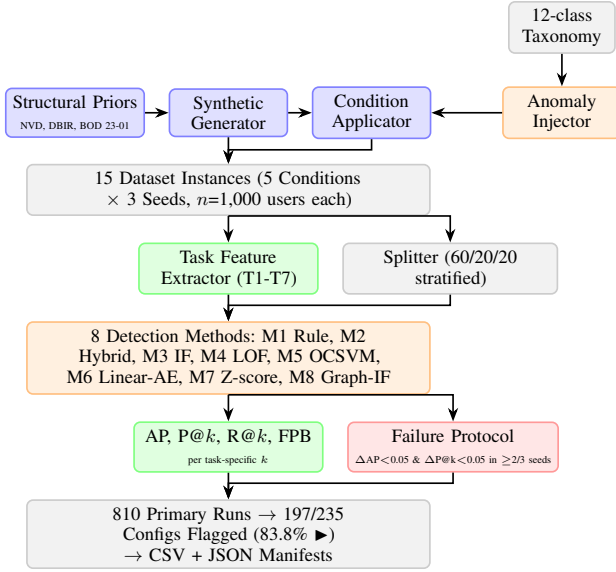


Fig. 1. HYGIENE BENCH end-to-end pipeline. Shaded boxes indicate the generator/condition layer (blue), evaluation harness (green), injector (orange), and outputs (gray/red).

III. HYGIENE BENCH DESIGN

A. System Architecture

Fig. 1 shows the HYGIENE BENCH pipeline. A seeded generator produces eleven entity/event tables under configurable telemetry conditions. An anomaly injector marks ground-truth labels across twelve anomaly classes. A deterministic splitter partitions entities into train/val/test splits. The evaluation harness extracts task-specific features, fits eight detection methods, and applies the pre-registered failure protocol to produce primary results and failure flags.

B. Schema and Generator

HYGIENE BENCH v0.1 generates eleven entity/event tables and one anomaly label table (Table I). The generator uses `numpy.random.default_rng(seed)` for all randomness; entity IDs are deterministic UUIDs from `hashlib.md5(seed_tag + index)`.

Structural priors. Patch-lag distributions follow DBIR 2026 aggregate statistics (43-day mean for critical patches). CVE severity distributions follow NVD base score bins. Telemetry cadence thresholds follow CISA BOD 23-01 (14-day asset discovery, 72-hour patch data) [1]. AD group-size distributions have no peer-reviewed source and are modeled via disclosed assumptions (power-law approximation) documented in the dataset card.

Conditions. Five parameterized conditions isolate the effect of telemetry quality on detection: **C-BASE** (no staleness); **C-FRESH** (elevated freshness flagging); **C-STALE** (20% of entities have >14-day telemetry gaps); **C-MISS** (one data source absent for 15% of entities); **C-UNSUP** (C-BASE with anomaly labels withheld from training). Each condition is generated for seeds 42, 137, and 2024, yielding 15 dataset instances.

TABLE I
HYGIENE BENCH V0.1 SCHEMA (MEDIUM SCALE, $n=1,000$ USERS)

Table	Rows	Description
users	1,000	AD user accounts, identity state
groups	65	AD security/distribution groups
computers	830	Managed endpoints
assets	830	Asset inventory entries
login_events	~249k	Authentication events (30 days)
group_membership_events	~51	Group add/remove events
account_lifecycle_events	varies	Account create/disable/enable
endpoint_patch_state	830	Patch compliance snapshot
vulnerability_records	~3.5k	Per-endpoint CVE exposures
remediation_events	varies	Patch/remediation actions
telemetry_freshness_log	varies	Per-entity source freshness
anomaly_labels	110	Ground truth (task + split)

TABLE II
CYBER-HYGIENE ANOMALY TAXONOMY. ATT&CK MAPPINGS ARE ENABLING-CONDITION CORRELATIONS ONLY.

Code	Class	ATT&CK enabling condition	Task(s)
AH-01	Stale privileged account	Valid Accounts (T1078.002)	T1
AH-02	Privilege escalation drift	Domain Policy Modification	T7
AH-03	Group membership drift	Domain Policy Mod. (T1484)	T2
AH-04	Dormant account reactivation	Valid Accounts (T1078.002)	T6
AH-05	Impossible/unusual login	Lateral Movement enabling	T1,T3
AH-06	Endpoint-identity risk corr.	Initial Access enabling	T3
AH-07	Patch noncompliance cluster	Exploit Public-Facing (T1190)	T5
AH-08	KEV exposure aging	Exploit Public-Facing (T1190)	T5
AH-09	Asset inventory mismatch	Defense Evasion enabling	T4
AH-10	Missing endpoint agent	Defense Evasion enabling	T4
AH-11	Telemetry missingness cluster	Defense Evasion enabling	T4
AH-12	Abnormal remediation delay	Exploit Public-Facing	T5

C. Anomaly Taxonomy

Table II lists the twelve anomaly classes with ATT&CK enabling-condition mappings. These mappings denote correlated enabling conditions, not detection claims for specific techniques.

D. Benchmark Tasks

Seven tasks cover the twelve anomaly classes across different entity scopes and temporal windows (Table III). The k values in Table III are pre-registered per TASK_SPECS.md and reflect operational SOC daily review budgets, not post-hoc tuning.

IV. EXPERIMENTAL SETUP

A. Detection Methods

Table IV describes the eight methods. All methods are applied to pre-computed task feature matrices; missing values are imputed with training-set column means. M8 is excluded

TABLE III
BENCHMARK TASKS. IMBALANCE RATIOS ARE PER DATASET INSTANCE
($n=1,000$).

Task	Description	Scope	k	Imbalance
T1	Stale privileged account risk	Priv. users	10	1:50
T2	Group membership drift	Users w/ grp events	20	1:167
T3	Endpoint-identity risk correlation	Users w/ endpoints	15	1:333
T4	Telemetry coverage gaps	All computers/assets	20	1:14
T5	Patch-vulnerability hygiene	All computers	25	1:59
T6	Dormant account reactivation	Users w/ lifecycle	10	1:250
T7	Escalation drift detection	Users w/ grp adds	10	1:500

TABLE IV
DETECTION METHODS EVALUATED IN HYGIENEBENCH

ID	Name	Type	Notes
M1	Rule baseline	Task rules	Per-task thresholds; no training
M2	Hybrid scorer	Weighted sum	Identity + endpoint + vuln + freshness
M3	Isolation Forest	Ensemble	200 trees, contamination=0.02
M4	LOF	Density	$k=20$, novelty mode, MinMaxScaler
M5	OCSVM	Kernel	$\nu=0.05$, RBF, MinMaxScaler
M6	Linear-AE	Reconstruction	PCA-based (dim=8); linear AE approx.
M7	Temporal z-score	Statistical	Population z-score on task features
M8	Graph-IF	Graph+ensemble	Bipartite networkx + IF; DOMINANT approx. [8]

from T4 and T5 by design (no meaningful graph signal for asset/patch anomalies).

B. Feature Engineering

For each task, a deterministic feature extractor computes a flat tabular feature matrix from the eleven raw CSV tables. Features are task-scoped: only columns relevant to the task entity type and anomaly class are included. Table V lists the primary feature groups per task.

Missing values (arising under C-MISS) are imputed with training-set column means before fitting. All features are computed identically for every method; no method-specific feature selection or preprocessing is applied beyond MinMaxScaler where noted in Table IV.

C. Metrics and Failure Protocol

Primary: Average Precision (AP) and Precision@ k ($P@k$) at the task-specific k values in Table III. **Secondary:** Recall@ k , False Positive Burden (FPB = $1 - P@k$), and AP rank stability ($1 - CV$ across seeds).

Pre-registered failure protocol. A (condition, task, method) triple is flagged as a *negative result* (►) when:

$$\begin{aligned} &AP_{\text{method}} - AP_{\text{rule}} < 0.05 \\ &\text{AND } P@k_{\text{method}} - P@k_{\text{rule}} < 0.05 \end{aligned} \quad (1)$$

in at least $\lceil \frac{2}{3} \times 3 \rceil = 2$ of 3 seeds. M1 (rule) is the reference; it is not flagged. The protocol was locked in the project decision log prior to any results being examined.

TABLE V
PRIMARY TASK FEATURES (3-5 PER TASK). ALL FEATURES COMPUTED FROM RAW TABLES; NO FEATURES CONSTRUCTED POST-HOC.

Task	Key features
T1	days_since_last_logon, privileged_group_count, is_service_account, source_freshness_flag
T2	group_change_count_30d, priv_group_change_count_30d, priv_add_ratio, actor_diversity
T3	is_privileged, days_since_agent_heartbeat, patch_compliance_score, open_kev_count, network_segment_risk
T4	endpoint_agent_installed, inventory_mismatch_flag, days_since_heartbeat, freshness_gap_days
T5	patch_compliance_score, critical_missing_patch_count, open_kev_count, max_kev_days_open, remediation_latency_mean
T6	dormant_days_at_reactivation, off_hours_rate, cross_segment_rate, is_privileged
T7	priv_adds_30d, priv_add_rate, group_change_velocity, actor_count

TABLE VI
MEAN AP AND $P@k$ ACROSS ALL 810 RUNS. ► = FAILURE-FLAGGED.
†M8 COVERS T1-T3, T6-T7 ONLY ($n=25$ CONFIGS).

Method	AP	$P@k$	Failure rate
M1 Rule baseline	0.916	0.444	,
M5 OCSVM	0.913	0.447	68.6% ►
M8 Graph-IF†	0.801	0.348	100% ►
M4 LOF	0.800	0.445	71.4% ►
M6 Linear-AE	0.798	0.407	85.7% ►
M3 Isolation Forest	0.781	0.428	85.7% ►
M7 Temporal z-score	0.774	0.392	85.7% ►
M2 Hybrid scorer	0.709	0.417	94.3% ►

V. RESULTS

A. Overall Performance

Table VI summarizes performance across all 810 runs. **197 of 235 (condition, task, method) configurations are failure-flagged (83.8%)**. The rule baseline (M1) achieves the highest mean AP (0.916). M5 (OCSVM) is the strongest ML method ($AP = 0.913$, failure rate 68.6%). M8 (graph) is failure-flagged on every configuration (100%). An earlier pipeline version aggregated failure flags per (condition, seed) instance; all numbers reported here use the pre-registered per-condition aggregation across the three seeds.

Figure 2 shows the per-(method, task) AP heatmap averaged across all conditions and seeds. The high failure rate is specific to synthetic, structured, class-imbalanced tabular anomaly detection. Anomaly injection creates features directly correlated with anomaly class, which rule thresholds exploit cleanly. Operational environments involve noisier, less discriminating feature distributions; HYGIENEBENCH should be read as a lower-difficulty bound.

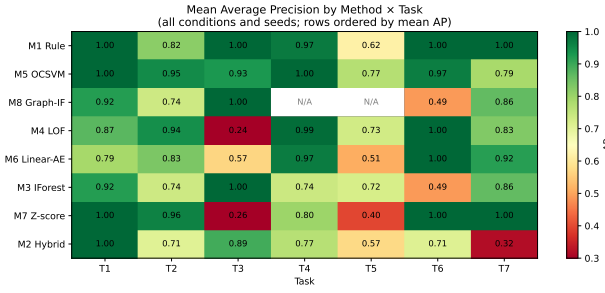


Fig. 2. Mean AP by method and task, averaged across all five telemetry conditions and three seeds. Darker cells indicate higher AP. M1 (rule baseline) dominates on 5 of 7 tasks; M5 (OCSVM) and M7 (temporal z-score) achieve the highest ML AP on T2 and T5 respectively.

TABLE VII
CONFIGURATIONS WHERE ML CONSISTENTLY OUTPERFORMS RULE BASELINE (B/S = BASE + STALE CONDITIONS COMBINED).

Task	Method	Rule AP	ML AP	Δ AP	Cond.
T2	M7 Z-score	0.766	0.951	+0.185	B/S
T2	M4 LOF	0.855	0.975	+0.120	C-MISS
T5	M5 OCSVM	0.458	0.668	+0.210	B/S
T5	M5 OCSVM	0.726	0.837	+0.111	C-MISS

B. Where ML Adds Value

ML consistently outperforms the rule baseline on two tasks (Table VII; Figure 3 shows full per-condition breakdowns).

T2, Group Membership Drift. The rule baseline achieves AP = 0.766 (C-BASE). Three ML methods consistently improve on this: M7 temporal z-score (AP = 0.951, Δ AP = +0.185), M5 OCSVM (AP = 0.913, +0.147), and M4 LOF (AP = 0.890, +0.124). Group-membership anomalies involve multi-dimensional population-level deviations that temporal z-score and kernel-based methods capture more precisely than fixed-threshold rules.

T5, Patch-Vulnerability Hygiene. Rule AP = 0.458 (C-BASE). Best ML: M5 OCSVM (AP = 0.668, Δ AP = +0.210). T5 combines multiple vulnerability signals (CVSS score, KEV exposure, remediation latency, days open) in ways that benefit from learned combinations. Absolute AP remains low across all methods, indicating genuine task difficulty.

C. Telemetry Condition Effects

Figure 4 plots the AP distributions by condition. C-STALE produces the lowest mean AP (0.741 vs. C-BASE 0.772, Δ = -0.031). The largest staleness degradation is T3 (endpoint-identity correlation): AP drops from 0.782 (C-BASE) to 0.613 (C-STALE, Δ = -0.169). T3 combines identity-side and endpoint-side features; staleness in either degrades the joint signal, making T3 the strongest argument for staleness-aware detection methods.

Table VIII summarizes mean AP by condition. C-FRESH, C-MISS, and C-UNSUP show higher mean AP than C-BASE (0.856, 0.845, 0.844 respectively). The primary contributor is T5, which gains +0.238 AP under C-FRESH. Fresh telemetry

TABLE VIII
MEAN AP BY TELEMETRY CONDITION (ALL METHODS AND TASKS).

Condition	Mean AP	Δ vs C-BASE	Primary driver
C-FRESH	0.856	+0.084	T5 +0.238, T1 +0.101
C-MISS	0.845	+0.073	Similar to C-FRESH
C-UNSUP	0.844	+0.072	Instance-level sampling
C-BASE	0.772		Reference
C-STALE	0.741	-0.031	T3 -0.169

improves patch compliance scores and KEV exposure count reliability, making underlying feature signals more discriminating, a data-quality effect, not a freshness-flag artifact. The C-UNSUP gap deserves an honest caveat: all eight methods are unsupervised, so withholding labels cannot raise AP. The C-UNSUP/C-BASE difference instead reflects independent RNG draws across separately generated dataset instances. Consequently, between-condition gaps of similar magnitude, including part of the C-FRESH effect, should be read with instance-level sampling variation in mind.

D. Failure-Flag Analysis

Figure 5 shows the per-(method, task) failure-flag heatmap. M8 (graph-augmented IF) is failure-flagged on all 25 applicable configurations (100%). The largest M8 deficit is T6 (dormant account reactivation): M8 AP = 0.490 vs. M1 AP = 1.000. Bipartite user-group graph features capture structural position but not temporal account state, which is the primary T6 signal. M5 (OCSVM) has the lowest failure rate (68.6%) and is not flagged on T2 or T5, the two tasks where ML adds value (it does fail Eq. (1) on T2 in occasional individual seeds, but never in $\geq 2/3$ seeds). M2 (hybrid scorer) has the highest failure rate among non-graph methods (94.3%), indicating that its fixed-weight combination is not well-calibrated across the heterogeneous task set.

T7 (escalation drift) is rule-dominated despite extreme rarity: with only 2 anomalous entities per dataset instance (imbalance $\approx 1:500$), the rule-based methods M1 and M7 achieve AP = 1.000, while several ML methods rank the two rare anomalies poorly, M2 has mean T7 AP = 0.319, M5 0.786, M4 0.830, and M3/M8 0.861 (task mean 0.822). No ML method closes the gap to the rule baseline.

VI. DISCUSSION AND LIMITATIONS

Interpretation. The 83.8% failure rate is not a finding against ML in security operations broadly. It is a finding about unsupervised ML vs. structured rule baselines on *synthetic, class-imbalanced tabular anomaly detection* in the hygiene domain. Real operational environments involve noisier data, absent labels, and anomaly presentations more overlapping with normal behavior. The synthetic benchmark tests whether methods can detect structurally clear anomalies; the two tasks where ML adds value (T2, T5) are those involving multi-signal combinations that benefit from learned representations.

Limitations. (1) All telemetry is synthetic; performance on real operational data will differ. (2) M6 is implemented as

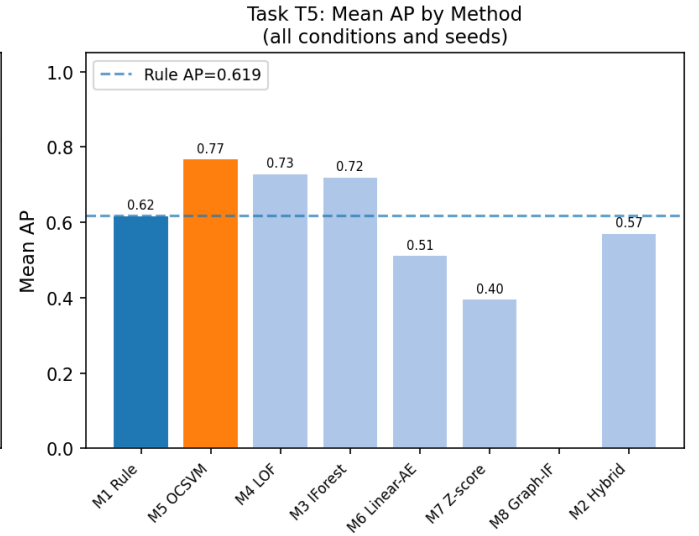
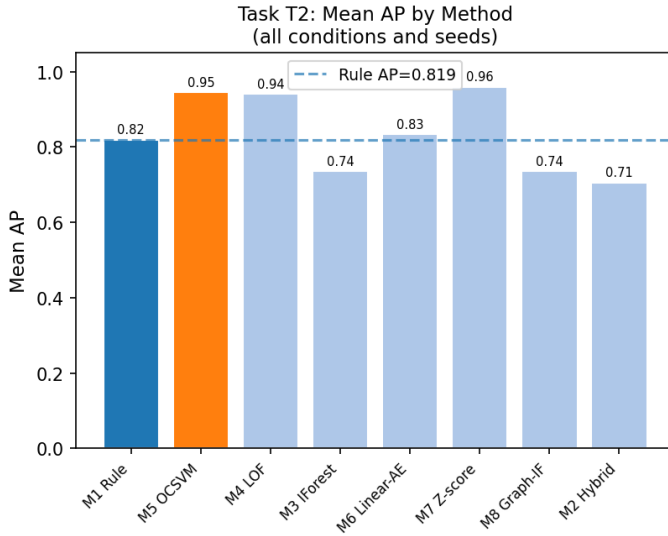


Fig. 3. T2 (group membership drift) and T5 (patch/vulnerability hygiene) AP by method and condition. These are the only two tasks where ML consistently outperforms the rule baseline by the pre-registered $\Delta \geq 0.05$ threshold in $\geq 2/3$ seeds. M7 (temporal z-score) achieves the best AP on T2 (+0.185 over rule); M5 (OCSVM) achieves the best AP on T5 (+0.210 over rule).

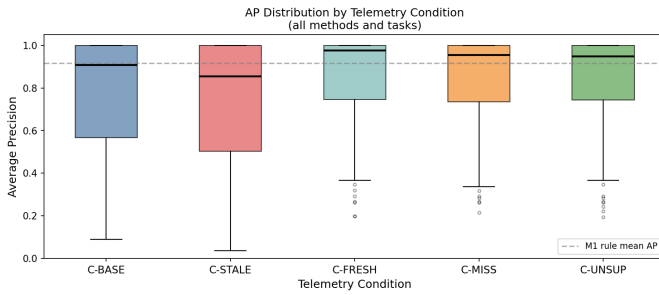


Fig. 4. AP distribution by telemetry condition across all methods and tasks. C-STALE produces the lowest median AP; C-FRESH and C-MISS elevate performance on tasks where telemetry freshness is discriminative (primarily T1 and T5).

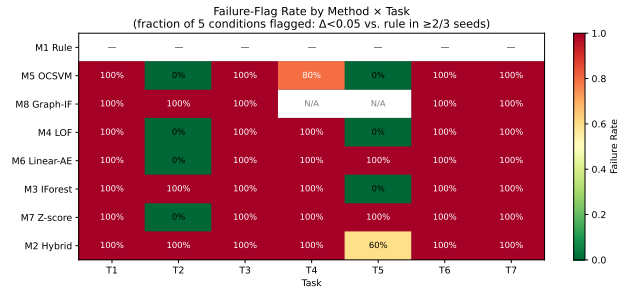


Fig. 5. Failure-flag fraction per (method, task) cell across all conditions and seeds. Red = high failure rate ($\geq 80\%$); blue = low failure rate. M8 (graph-IF) is uniformly red (100% failure rate). M5 (OCSVM) achieves the lowest failure rate (68.6%).

PCA-based linear reconstruction error, not a neural autoencoder; M8 uses networkx features rather than a full deep graph autoencoder [8]. Both are disclosed approximations. (3) Population scale ($n=1,000$) is smaller than most public-

sector environments. (4) Feature extractors encode domain knowledge; methods exploiting these features appear better. (5) No score calibration is performed; calibration is excluded by design. (6) AD group-size distributions lack a peer-reviewed prior source.

Practical implications. HYGIENEBENCH supports three practitioner decisions. First, for tasks where rule baselines already achieve $AP \geq 0.90$ (T1, T4, T6), practitioners should weigh the maintenance overhead of ML against minimal detection gain: HYGIENEBENCH provides no evidence that ML justifies that overhead on synthetic structured data. Second, for T2 and T5, ML (OCSVM, temporal z-score, LOF) provides measurable lift over rules, motivating investment in supervised or semi-supervised variants if labels become available. Third, the C-STALE results quantify the detection cost of stale data pipelines: the -0.169 AP drop on T3 under C-STALE is a conservative lower bound on what SOC's risk when telemetry ingestion lags.

Future work. Planned extensions include: (i) $n=10,000$ large-scale instances to test statistical power under lower imbalance ratios; (ii) supervised baselines (gradient-boosted trees) to bound the label-value ceiling; (iii) full neural autoencoder (M6) and DOMINANT-style deep graph autoencoder (M8) once a GPU environment is available; (iv) PoC-label and ExploitDB-enriched vulnerability records to make T5 harder; (v) catalog-perturbation sensitivity sweeps on the structural priors (DBIR patch lag, NVD severity) to characterize robustness to prior misspecification.

VII. CONCLUSION

We introduced HYGIENEBENCH, a reproducible synthetic benchmark for cyber-hygiene anomaly detection spanning seven tasks, twelve anomaly classes, and eleven entity/event tables generated from a fully open, seeded pipeline grounded

in public structural priors (NIST NVD, Verizon DBIR 2026, CISA BOD 23-01 [1]). An 810-run evaluation of eight methods under five telemetry conditions, using a pre-registered failure protocol, finds that simple rule baselines perform at least as well as unsupervised ML on **83.8%** of (condition, task, method) configurations. ML adds consistent value on group membership drift (T2, +0.185 AP, M7 temporal z-score) and patch-vulnerability hygiene (T5, +0.210 AP, M5 OCSVM). M8 (graph-IF) is failure-flagged on all 25 applicable configurations. Telemetry staleness consistently degrades performance; T3 (endpoint-identity correlation) is the most sensitive task (−0.169 AP under C-STALE), quantifying the detection cost of stale data pipelines.

These results motivate three benchmark-practice changes: (1) task-stratified reporting over macro-averaged summaries; (2) negative-result protocols as first-class contributions; (3) telemetry freshness as a mandatory evaluation axis.

HYGIENEBENCH v0.1 (generator, datasets, evaluation harness) is openly released at <https://doi.org/10.5281/zenodo.20438401>.

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